

Prarthana Shiwakoti

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EDUCATION

Fordham University – New York, NY | **M.S. Computer Science**

Aug 2024 – May 2026

Graduate Research Assistant | **GPA: 4.0**

Relevant Coursework: Programming Languages, Mobile Device Programming, Cloud Computing, Software Engineering

Vellore Institute of Technology (VIT) – B.Tech Computer Science & Engineering

Jun 2020 – May 2024

Full tuition waiver | **GPA: 9.34/10**

Relevant Coursework: Parallel and Distributed Computing, Web Programming, Databases, Operating Systems, Algorithms

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C++ , Swift, SQL,Go

Frameworks & Libraries: React, Node.js, Express.js, Next.js, Django, Scrapy, Flask, PyTorch, Transformers, LangChain

Databases: PostgreSQL, MySQL, MongoDB, SQLite, BigQuery

Tools & Technologies: Docker, Kubernetes, Git, Google Cloud Platform, Azure DevOps, Figma, Tableau

WORK EXPERIENCE

AI Researcher

Aug 2024 – Present

Fordham University – New York, NY

- Worked on research focused on **Hate Speech Detection** in **Indic languages** using the **BERT language model**, aiming to enhance the accuracy and efficiency of identifying hate speech in under represented regional languages.

AI/ML Intern

Jun 2025 – Present

Andor Health – Orlando, FL (On-site)

- Developed **AI-powered Ambient Nursing Documentation System** combining **Whisper speech-to-text** with **GPT-4o Realtime**, automating form completion through structured prompt engineering and deterministic model configuration, achieving **18 minutes saved per patient admission** and **40% operational efficiency improvement**.
- Implemented **fall/intrusion detection algorithms** for **Virtual Patient Monitoring** through comparative evaluation of **MoveNet**, **FaceMesh**, and **FastestDet models**, with performance optimization using **VitPose+** and pose regression techniques, integrated with **COCO-SSD** for comprehensive behavioral analysis within ICU environments, achieving **51% decrease in patient fall rates**, and **70% reduction in sitter costs**.

Software Developer Intern

Dec 2023 – Jul 2024

Resimplifi Inc. — South Carolina, USA (Remote)

- Executed **Python Scrapy** spiders to meticulously extract crucial data from US real-estate sites, integrating it into **Postgres** from the scraped pool, resulting in a notable **20% reduction in query time**.
- Developed and tailored 20+ REST APIs with Django REST Framework, leveraging PostgreSQL for storage and MongoDB for caching, optimizing real-time access to the platform.
- Designed and deployed automated ETL workflows incorporating LLM prompting to intelligently extract and standardize key property features from unstructured text across one million plus listing, boosting data accuracy by 20% and eliminating 40+ hours/week of manual data tagging.

RECENT PROJECTS AND PUBLICATIONS

CareerScape (Unified Job Board)

- Created a full-stack platform using **Swift** for the mobile app, **Go** for backend and API development, and **PostgreSQL** for the database for a comprehensive job board solution to streamline the application process for job seekers.
- Automated data scraping with **Scrapy** and **Selenium** from multiple job boards, streamlining the collection of over **200 job listings daily** and eliminating the need for manual searches, enhancing user convenience and efficiency.
- Executed the development of an admin panel that supports continuous, real-time scraping, ensuring the platform is always up-to-date with the latest job opportunities and providing users with timely and relevant job listings.

Multimodal RAG Pipeline Framework for Productivity Assessment AI Assistant

- Engineered end-to-end **ETL pipeline** extracting and staging **Azure DevOps** data (Git commits, user stories, sprint metrics, pipeline logs) into **BigQuery data warehouse**, performing comprehensive feature extraction and exploration analysis using **GCP Vertex AI**.
- Implemented hybrid **RAG+TAG analytics architecture** conducting quantitative developer productivity analysis (commit volume, diff entropy, sprint velocity) through **Table-Augmented Generation (TAG)** using **Gemini 2.0 Flash** and qualitative assessment of ticket discussions, code reviews, and team communications via **Retrieval-Augmented Generation** using **Qwen-7B LLM** with **bge-large-en embeddings** and **FAISS vector databases**.

Sentiment Analysis for Product Reviews using Long Short-Term Memory with Layer Normalization

- Published in **IEEE**, this paper proposes sentiment analysis on customer reviews using a novel **LSTM with layer normalization** technique. Achieved superior accuracy in **LSTM with layer normalization** model compared to existing ones.